Coprime Ramsey Numbers: Exact Values and Structural Properties

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Research draft, corrected October 17, 2026

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Abstract

We introduce the *coprime Ramsey number* $R_{\text{cop}}(k)$, defined as the minimum n such that every 2-coloring of the edges of the coprime graph on $\{1, \ldots, n\}$ contains a monochromatic copy of K_k. Combining SAT-computed lower bounds with an elementary clique upper bound, we establish $R_{\text{cop}}(2) = 2$, $R_{\text{cop}}(3) = 11$, and $R_{\text{cop}}(4) = 59$, compute the 3-color triangle number $R_{\text{cop}}(3;3) = 53$, and determine exact values for path, cycle, asymmetric, and Gallai variants. All clique-type values are prime. This observation, first recorded here as a conjecture, has since been proved in general by Du, Xi, Deng, and Ma, who showed $R_{\text{cop}}(k) = p_{R(k,k)-1}$, the $(R(k,k)-1)$-th prime, for the edge-coloring variant studied in this paper. Their complete-pattern transfer also covers Gallai triangles. Paths and cycles require separate lower-bound arguments; the two longest path values remain historical computational reports awaiting independent reproduction.

Mathematics Subject Classifications: 05D10, 05C55, 11A05

Keywords: Ramsey theory, coprime graph, SAT solving, graph coloring.

1 Introduction

The *coprime graph* $G_{\text{cop}}(n)$ has vertex set $[n] = \{1, 2, \ldots, n\}$ with $\{i, j\}$ an edge whenever $\gcd(i, j) = 1$. This graph, studied by Erdős and Sárközy [3] and others [6, 1], encodes number-theoretic structure in a combinatorial object: its edge density converges to $6/\pi^2 \approx 0.608$, its clique number equals $1 + \pi(n)$ (where π is the prime-counting function), and its independence number is $\lfloor n/2 \rfloor$ for $n \geqslant 2$ (see Section 4 for short proofs).

We propose a Ramsey-theoretic study of this graph. Classical Ramsey numbers $R(k, k)$ [7] concern the complete graph K_n; replacing K_n with $G_{\text{cop}}(n)$ yields a problem where Ramsey-type forcing must contend with the arithmetic structure of coprimality.

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The values reported here were computed and announced first (March 2026, in the repository accompanying this paper). Subsequently Du, Xi, Deng, and Ma [2], citing those computations, proved a closed form for the clique-type numbers, $R_{\mathrm{cop}}(k) = p_{R(k,k)-1}$; Remark 10 details the correspondence. Their Proposition 4.6 also covers Gallai triangles. The path and cycle variants of Section 3 have non-complete targets, for which the prime-clique upper bound remains available but the matching lower-bound transfer can fail.

Definition 1. The *coprime Ramsey number* $R_{\mathrm{cop}}(k)$ is the minimum n such that every 2-coloring of $E(G_{\mathrm{cop}}(n))$ contains a monochromatic K_k. More generally, $R_{\mathrm{cop}}(k; c)$ is the minimum n such that every c-coloring of $E(G_{\mathrm{cop}}(n))$ contains a monochromatic K_k (so $R_{\mathrm{cop}}(k) = R_{\mathrm{cop}}(k; 2)$).

Since $E(G_{\mathrm{cop}}(n)) \subseteq E(K_n)$, the restriction of any K_k-avoiding coloring of $E(K_n)$ to the coprime edges is an avoiding coloring of $G_{\mathrm{cop}}(n)$: every clique of $G_{\mathrm{cop}}(n)$ is a clique of K_n with the same edge colors. Taking $n = R(k,k) - 1$ gives $R_{\mathrm{cop}}(k) \geqslant R(k,k)$; forcing monochromatic cliques in the sparser coprime graph never requires fewer vertices, and empirically requires many more for $k \geqslant 3$. The same restriction argument shows that an avoiding coloring of $G_{\mathrm{cop}}(n)$ restricts to one of $G_{\mathrm{cop}}(m)$ for every $m \leqslant n$, so the forcing property is upward closed in n and each R_{cop} value is determined by the single SAT-to-UNSAT transition.

In the other direction, an elementary clique bound shows $R_{\mathrm{cop}}(k)$ is well defined and pins it to within a prime:

Proposition 2 (Clique upper bound)**.** $R_{\mathrm{cop}}(k) \leqslant p_{R(k,k)-1}$*, where* p_m *denotes the* m*-th prime. More generally, for* c *colors and targets* $k_1, \ldots, k_c$*, the corresponding coprime Ramsey number is at most* $p_{R(k_1,\ldots,k_c)-1}$*.*

Proof. Let $n = p_{R(k,k)-1}$, so that $\pi(n) = R(k,k) - 1$. The set $\{1\} \cup \{p \leqslant n : p \text{ prime}\}$ is pairwise coprime, hence a clique of $G_{\mathrm{cop}}(n)$ of size $1 + \pi(n) = R(k,k)$. Any 2-coloring of $E(G_{\mathrm{cop}}(n))$ restricts to a 2-coloring of the edges of this $K_{R(k,k)}$, which contains a monochromatic K_k by the definition of $R(k,k)$. The multicolor and asymmetric cases are identical. □

2 Exact values

Theorem 3. $R_{\mathrm{cop}}(2) = 2$.

Proof. Since $\gcd(1,2) = 1$, the graph $G_{\mathrm{cop}}(2)$ has a single edge, and any 2-coloring assigns it one color, producing a monochromatic K_2; while $G_{\mathrm{cop}}(1)$ has no edges, so no monochromatic K_2 is forced at $n = 1$. □

Theorem 4. $R_{\mathrm{cop}}(3) = 11$.

Proof. By exhaustive incremental extension. Among all 2^{21} edge 2-colorings of $G_{\mathrm{cop}}(8)$ (21 coprime edges), exactly 36 avoid monochromatic triangles. We extend these through

vertices 9, 10, 11: for each new vertex v, the coprime edges from v to $\{1, \ldots, v-1\}$ are enumerated, and every 2-coloring of these new edges is tested against each surviving avoiding coloring.

n	Coprime edges	New edges at n	Avoiding colorings
8	21	—	36
9	27	6	36
10	31	4	156
11	41	10	0

The increase from 36 to 156 at $n = 10$ occurs because $10 = 2 \times 5$ is coprime to only 4 predecessors, imposing weak constraints. The collapse to 0 at $n = 11$ is driven by 11 being prime: it is coprime to all of $\{1, \ldots, 10\}$, creating 10 new edges and far too many triangles for any 2-coloring to avoid. □

Remark 5. $R_{\text{cop}}(3)/R(3,3) = 11/6 \approx 1.83$. The excess over $R(3,3)$ reflects multiplicative correlations among missing edges: multiples of a prime p form correlated non-adjacencies in $G_{\text{cop}}(n)$ [3].

Theorem 6. $R_{\text{cop}}(4) = 59$.

Proof. Lower bound: for every $n \leqslant 58$, the Glucose4 solver [4] finds a 2-coloring of $E(G_{\text{cop}}(n))$ with no monochromatic K_4; by the restriction argument of Section 1 this gives $R_{\text{cop}}(4) \geqslant 59$. Upper bound: $R(4,4) = 18$, and $59 = p_{17}$ is the 17-th prime, so Proposition 2 gives $R_{\text{cop}}(4) \leqslant p_{17} = 59$. □

Remark 7 (Computational corroboration). Prior to the observation of Proposition 2, the upper bound was explored using sampled extensions: 100 distinct avoiding colorings of $E(G_{\text{cop}}(58))$ were extracted via SAT with blocking clauses, and for each one the instance encoding the extension to $n = 59$ (coloring the 58 new edges from the prime vertex 59) is UNSAT. The full $n = 59$ instance (1,085 variables, ∼118,000 clauses) did not terminate within 30 minutes under CDCL solving, which is why the seeded extensions were used; these samples do not exhaust the avoiding colorings at 58 and hence do not prove the upper bound. Proposition 2 supplies the complete upper-bound proof.

k	$R_{\text{cop}}(k)$	$R(k,k)$
2	2	2
3	11	6
4	59	18

Theorem 8. $R_{\text{cop}}(3;3) = 53$.

Proof. A full 3-coloring of $E(G_{\mathrm{cop}}(52))$ avoiding monochromatic triangles gives the lower bound; a new witness was independently checked against all 6,684 coprime triangles in the October 17, 2026 audit. For the upper bound, $R(3,3,3) = 17$ and Proposition 2 give $R_{\mathrm{cop}}(3;3) \leqslant p_{16} = 53$. The historical computation reported a direct UNSAT result at 53; that full search was not rerun in the audit and is not needed for this upper-bound proof. □

Conjecture 9 (Primality; now a theorem of Du–Xi–Deng–Ma [2])**.** All diagonal coprime Ramsey numbers $R_{\mathrm{cop}}(k)$ with $k \geqslant 2$ are prime. More generally, $R_{\mathrm{cop}}(k;c)$ is prime for all $k, c \geqslant 2$.

Evidence. $R_{\mathrm{cop}}(2) = 2$, $R_{\mathrm{cop}}(3) = 11$, $R_{\mathrm{cop}}(4) = 59$, and $R_{\mathrm{cop}}(3;3) = 53$ are all prime. The heuristic explanation is that prime values of n maximize the coprime density of $G_{\mathrm{cop}}(n)$—a prime p is coprime to all of $\{1, \ldots, p-1\}$—so the transition from SAT to UNSAT preferentially occurs at primes, where the addition of a universal coprime vertex creates the maximal number of new constraints. This conjecture has since been proved by Du, Xi, Deng, and Ma [2]; see Remark 10 below.

Remark 10 (Subsequent theoretical confirmation). The exact values introduced and computed above have since been given a closed-form explanation. Du, Xi, Deng, and Ma [2] study coprime Ramsey numbers in two formulations, a vertex-coloring and an edge-coloring version. The numbers $R_{\mathrm{cop}}(k)$ of the present paper, defined by 2-colorings of $E(G_{\mathrm{cop}}(n))$, are the *edge*-coloring version, and for it Du et al. prove (their Theorem 4.4) the prime-index identity

$$R_{\mathrm{cop}}(k) = p_{R(k,k)-1},$$

where p_m denotes the m-th prime and $R(k,k)$ the classical diagonal Ramsey number; the upper-bound half is exactly Proposition 2, and their contribution is the matching lower-bound construction. Their vertex-coloring version instead satisfies the elementary identity $p_{\sum_i (k_i-1)}$ (their Theorem 3.2), a different quantity. The values computed in this paper are exactly the values the identity predicts: with $R(3,3) = 6$ and $R(4,4) = 18$ one obtains $p_5 = 11$ and $p_{17} = 59$. Our asymmetric and multicolor values are likewise consistent with the mixed identity $p_{R(k_1,\ldots,k_c)-1}$ of their Theorem 4.4 in every computed case: $R_{\mathrm{cop}}(2,3) = 3 = p_2$ with $R(2,3) = 3$, $R_{\mathrm{cop}}(2,4) = 5 = p_3$ with $R(2,4) = 4$, $R_{\mathrm{cop}}(3,4) = 19 = p_8$ with $R(3,4) = 9$, and $R_{\mathrm{cop}}(3;3) = 53 = p_{16}$ with $R(3,3,3) = 17$. The primality observation of Conjecture 9 is thereby established unconditionally: $R_{\mathrm{cop}}(k)$ is a prime by construction. For the two-color diagonal, identifying the primes of the form $p_{R(k,k)-1}$ remains tied to the unknown classical values. In the unrestricted mixed family every prime occurs: $R(2,t) = t$ gives $R_{\mathrm{cop}}(2,t) = p_{t-1}$. Their Proposition 4.6 also transfers fixed color patterns on complete witnesses. In particular, monochromatic or rainbow triangles give $\mathrm{GR}_{\mathrm{cop}}(3;3) = p_{\mathrm{gr}_3(K_3)-1} = p_{10} = 29$, since $\mathrm{gr}_3(K_3) = 11$. Paths and cycles have non-complete targets; labels can coincide on non-adjacent vertices, so the matching pullback lower bound is unavailable in general. Their non-prime values (P_{cop} attains 9 and 10, and C_{cop} attains 8) demonstrate this boundary.

3 Ramsey variants on the coprime graph

We define and compute several Ramsey-type variants on $G_{\text{cop}}(n)$. All values are historical SAT computations. The October 17, 2026 audit reproduced the smaller exact transitions, but the path cases $k = 7, 8$ each exceeded a 60-second audit limit; those two reported values remain awaiting independent reproduction. Clique and Gallai triangle values also follow from the complete-pattern transfer.

Definition 11. Let $c \geqslant 2$ be the number of colors and $k \geqslant 2$ the target structure size.

1. $P_{\text{cop}}(k)$: minimum n such that every 2-coloring of $E(G_{\text{cop}}(n))$ contains a monochromatic path on k edges.

2. $C_{\text{cop}}(k)$: minimum n such that every 2-coloring of $E(G_{\text{cop}}(n))$ contains a monochromatic cycle on k edges.

3. $R_{\text{cop}}(s, t)$: minimum n such that every 2-coloring of $E(G_{\text{cop}}(n))$ has a monochromatic K_s in color 0 or K_t in color 1 (asymmetric coprime Ramsey).

4. $\text{GR}_{\text{cop}}(k; c)$: the minimum n such that every c-coloring of $E(G_{\text{cop}}(n))$ contains a monochromatic or rainbow K_k (the Gallai coprime Ramsey number; requires $c \geqslant 3$).

Table 1: Reported coprime Ramsey variants. The path values for $k = 7, 8$ await independent reproduction in the current audit.

Variant	**Values**	**Classical counterpart**
$P_{\text{cop}}(k)$, $k = 3, \ldots, 8$	5, 7, 9, 10, 13, 13	—
$C_{\text{cop}}(k)$, $k = 3, \ldots, 6$	11, 8, 13, 11	—
$R_{\text{cop}}(2, 3)$	3	$R(2, 3) = 3$
$R_{\text{cop}}(2, 4)$	5	$R(2, 4) = 4$
$R_{\text{cop}}(3, 4)$	19	$R(3, 4) = 9$
$R_{\text{cop}}(3; 3)$	53	$R(3, 3, 3) = 17$
$\text{GR}_{\text{cop}}(3; 3)$	29	$\text{gr}_3(K_3) = 11$

Remark 12. Several patterns emerge from Table 1.

1. *Path versus clique.* $P_{\text{cop}}(k) \ll R_{\text{cop}}(k)$: paths are much easier to force in the computed cases. The reported values plateau at 13 for $k = 7, 8$; that plateau has not yet been independently reproduced by the current audit.

2. *Even–odd cycle gap.* $C_{\text{cop}}(4) = 8 < C_{\text{cop}}(3) = 11$ and $C_{\text{cop}}(6) = 11 < C_{\text{cop}}(5) = 13$: even-length cycles are easier to force.

3. *Multi-color growth.* $R_{\mathrm{cop}}(3;3)/R_{\mathrm{cop}}(3;2) = 53/11 \approx 4.8$, compared with the classical $R(3,3,3)/R(3,3) = 17/6 \approx 2.8$. The coprime graph's structure amplifies multi-color avoidance.

4. *Coprime amplification.* For the clique-type variants with $k \geqslant 3$, the coprime version strictly exceeds its classical counterpart: $R_{\mathrm{cop}}(3) > R(3,3)$, $R_{\mathrm{cop}}(3;3) > R(3,3,3)$, $\mathrm{GR}_{\mathrm{cop}}(3;3) > \mathrm{gr}_3(K_3)$. (For $k = 2$ equality holds: $R_{\mathrm{cop}}(2) = 2 = R(2,2)$.)

4 Structural properties

For completeness we record short proofs of the structural parameters quoted in the introduction.

Proposition 13. $\omega(G_{\mathrm{cop}}(n)) = 1 + \pi(n)$.

Proof. The set $\{1\} \cup \{p \leqslant n : p \text{ prime}\}$ is a clique of size $1 + \pi(n)$. For the upper bound: in any pairwise coprime set $S \subseteq \{2, \ldots, n\}$, each element's prime factorization uses a disjoint set of primes, so $|S| \leqslant \pi(n)$. □

Proposition 14. *$G_{\mathrm{cop}}(n)$ is weakly perfect:* $\chi(G_{\mathrm{cop}}(n)) = \omega(G_{\mathrm{cop}}(n)) = 1 + \pi(n)$.

This follows from the observation that the partition

$$\{1\},\ \{2,4,6,\ldots\},\ \{3,9,15,\ldots\},\ \{5,25,35,\ldots\},\ \ldots$$

into classes indexed by the smallest prime factor gives a proper $(1+\pi(n))$-coloring. Vertex colorings and chromatic numbers of related coprime-type graphs on group elements were studied by Syarifudin and Wardhana [8].

Proposition 15. *For $n \geqslant 2$, $\alpha(G_{\mathrm{cop}}(n)) = \lfloor n/2 \rfloor$.*

Proof. The even numbers $\{2, 4, 6, \ldots, 2\lfloor n/2 \rfloor\}$ form an independent set of size $\lfloor n/2 \rfloor$ (every pair shares the factor 2). Conversely, let S be an independent set with $|S| \geqslant \lfloor n/2 \rfloor + 1 \geqslant 2$. Then $1 \notin S$, since 1 is adjacent to every other vertex, and S contains no two consecutive integers, since consecutive integers are coprime. But a subset of $\{2, \ldots, n\}$ with no two consecutive elements has size at most $\lceil (n-1)/2 \rceil = \lfloor n/2 \rfloor$, a contradiction. □

5 Methodology

All computations use the PySAT toolkit [5] with the Glucose4 [4] CDCL solver. The methodology varies by problem size.

Small two-color instances ($R_{\mathrm{cop}}(3)$, path and cycle variants). For each n, we encode the avoidance problem as a SAT instance: one Boolean variable per coprime edge ($x_e = \top$ for color 0, $x_e = \bot$ for color 1), two clauses per target substructure (forbidding all-color-0 and all-color-1). The solver returns SAT or UNSAT. The value $R_{\mathrm{cop}}(3) = 11$ is also verified by the exact incremental extension method of Theorem 4, which enumerates *all* avoiding colorings at each n and is thus a complete proof.

Large instances ($R_{\text{cop}}(4)$). The incremental SAT approach determines $R_{\text{cop}}(4) \geqslant 59$: Glucose4 returns SAT for all $n \leqslant 58$, with solving times under 1 second per instance. The upper bound is Proposition 2; the seeded extension experiment of Remark 7 provides additional corroboration.

Multi-color variants ($R_{\text{cop}}(3;3)$, $\text{GR}_{\text{cop}}(3;3)$). For c colors, each edge has $c-1$ Boolean variables with at-most-one pairwise exclusion constraints. For each clique and each color, a clause forbids monochromatic assignment. The Gallai variant additionally forbids rainbow cliques via all-distinct permutation clauses. These instances are within Glucose4's reach for the parameter ranges in Table 1.

Acknowledgements

Computational analysis was performed with the assistance of an AI system (Claude, Anthropic), used for exploring the problem space and generating candidate SAT encodings. Codex (OpenAI) assisted the October 17, 2026 audit and subsequent software and documentation repairs. The audit independently checked full lower-bound witnesses at 10, 58, and 52 and reproduced the smaller variant transitions; the two longest path reports remain qualified as described in Section 3. No Lean certification of the coprime Ramsey results is claimed. The author takes full responsibility for the correctness of the statements and proofs. The author thanks Terry Tao and the contributors to the Erdős problems database [9] for creating a resource that motivated this investigation.

Code and data are available at github.com/queelius/computational-explorations.

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